SheHarm-CAR: Counterfactual and Rule-Guided Neuro-Symbolic Reasoning for Women-Targeted Harmful Meme Detection

Abstract

Women-targeted harm in memes may be conveyed through sexual harassment, violence, misogyny, appearance attacks, and character assassination. Such harm may be explicit or may emerge implicitly from the interaction between image and text. We introduce **SheHarm-Meme**, a multi-task benchmark for explainable women-targeted harmful meme understanding. It supports four tasks: women-related target identification, three-way harmfulness classification, five-way harm-category classification, and evidence-grounded rationale generation. We further propose **SheHarm-CAR**, a counterfactual, target-guided, and rule-grounded neuro-symbolic framework that identifies the women-related target, retrieves target-relevant concepts from a Women-Harm Knowledge Ontology, and contrasts harm-supporting rules with contextual exception rules. Confidence-gated fusion integrates multimodal, ontological, and symbolic evidence, while neural–symbolic consistency and counterfactual learning encourage predictions and rationales to depend on relevant evidence. SHEHARM-CAR achieves 84.62 Harm-F1, 76.88 Joint-F1, 89.34 BERTScore, and 85.6 counterfactual faithfulness, outperforming the strongest baseline by 2.04 and 2.59 points in Harm-F1 and Joint-F1, respectively. These results suggest that target-conditioned ontology retrieval, contrastive rule reasoning, and counterfactual learning improve predictive performance and rationale faithfulness.

SheHarm-Meme Annotation Example

OCR Text: *Get back in the kitchen!*

Women-related Target	woman being struck
Harmfulness	*Explicit-Harm*
Harm Category	*Violence*
Rationale	The meme reinforces the stereotype that women belong in the kitchen while depicting aggression toward a woman.

Figure 1: SheHarm-Meme annotation for explicit misogyny conveyed through a gender-role stereotype and supporting visual aggression.

1 Introduction

Internet memes combine images and short text to communicate humour, opinions, and social commentary (Hee and Lee, 2025; Chen et al., 2025). The same format can also spread women-targeted harassment and discriminatory beliefs while appearing humorous or harmless (Lin et al., 2025). Such memes[1] may express sexual harassment, violence, misogyny, appearance attacks, and character assassination through direct insults, stereotypes, innuendo, sarcasm, or culturally grounded references (Kumari et al., 2025). Their harmfulness may arise from visual symbolism, background knowledge, or interactions between individually benign image and text components (Kiela et al., 2020; Kumar and Nandakumar, 2022; Burbi et al., 2023; Lin et al., 2023).

Most harmful-meme studies formulate the task as binary multimodal classification using feature fusion, cross-modal interaction, or multimodal pretraining (Kumar and Nandakumar, 2022; Burbi

[1] **Content warning:** This paper contains examples and discussion of women-targeted abusive, violent, misogynistic, and discriminatory content that some readers may find disturbing.

et al., 2023). More recent methods use large language or multimodal models to obtain external knowledge, reasoning traces, and natural-language explanations (Lin et al., 2023, 2025; Tzelepi and Mezaris, 2024). Although these approaches improve implicit-harm detection, a meme-level label does not identify the targeted woman or women-related entity, the form of harm, or the evidence supporting the decision. Multimodal misogyny benchmarks provide fine-grained labels for stereotyping, objectification, shaming, hostility, and violence (Fersini et al., 2022; Muti et al., 2022), together with explanation generation for women-targeted memes (Kanwar et al., 2025). However, explicit and implicit harms are often merged, and generated rationales may remain plausible without reflecting the evidence actually used by the model (Lin et al., 2023, 2025).

Implicit harm may additionally depend on social, cultural, and commonsense knowledge absent from the meme itself. Existing methods therefore incorporate LMM-generated descriptions or external resources such as ConceptNet and Wikidata (Tzelepi and Mezaris, 2024; Tripathi et al., 2026). Retrieved knowledge, however, may be unrelated to the actual target and does not by itself provide a transparent reasoning process. Neuro-symbolic learning offers a complementary direction by combining neural representations with structured knowledge and interpretable rules (Liu et al., 2025; Wang et al., 2025). This is particularly useful for distinguishing harmful endorsement from quotation, criticism, awareness, and counter-speech.

We introduce **SheHarm-Meme**, a multi-task benchmark for explainable women-targeted harmful meme understanding. It supports four tasks: *(i)* women-related target identification, *(ii)* three-way harmfulness classification into *Explicit-Harm*, *Implicit-Harm*, and *Non-Harm*, *(iii)* five-way harm-category classification, and *(iv)* evidence-grounded rationale generation. Harmful memes receive one of five categories: *Sexual-Harassment*, *Violence*, *Misogyny*, *Appearance-Attack*, or *Character-Assassination*; for *Non-Harm* memes, the category is set to *None* and excluded from category training and evaluation.

We further propose **SheHarm-CAR**, a counterfactual, target-guided, and rule-guided neuro-symbolic framework. It identifies the women-related target, retrieves target-relevant concepts from a Women-Harm Knowledge Ontology, contrasts harm-supporting rules with contextual exception rules, and integrates multimodal, ontological, and symbolic evidence through confidence-gated fusion. Neural–symbolic consistency aligns predictions with activated rules, while counterfactual interventions encourage harmfulness, category, and rationale outputs to depend on relevant evidence.

Our contributions are threefold:

- We introduce **SheHarm-Meme**, a four-task benchmark with ontology-linked women-related targets, explicit/implicit/non-harmful labels, five harm categories, and human-written rationales.
- We propose **SheHarm-CAR**, which combines target-conditioned ontology retrieval with contrastive harm-supporting and contextual exception rules.
- We develop confidence-gated neural–symbolic fusion, consistency regularization, and counterfactual learning, and evaluate their effects through main-task performance, implicit-harm analysis, ablations, and rationale-faithfulness measures.

Research Questions. We investigate four research questions:
RQ1: How accurately can models identify the women-related target and distinguish between explicit, implicit, and non-harmful memes?
RQ2: How effectively can harmful memes be classified into the five fine-grained harm categories?
RQ3: Does target-conditioned ontology retrieval and contrastive rule reasoning improve performance over standard multimodal models?
RQ4: Do neural–symbolic consistency and counterfactual learning improve the faithfulness and informativeness of generated rationales?

2 Related Work

Harmful and misogynistic memes. The Hateful Memes Challenge established meme harm detection as a multimodal task requiring joint image–text interpretation (Kiela et al., 2020). Later methods strengthened cross-modal interaction using CLIP, prompting, and textual inversion (Kumar and Nandakumar, 2022; Cao et al., 2022; Burbi et al., 2023). MAMI introduced binary and fine-grained misogyny labels, while subsequent work examined unintended bias and explanation generation (Fersini et al., 2022; Muti et al., 2022; Kumari et al., 2024; Kanwar et al., 2025). However, these studies generally operate at the meme level and do not jointly identify the targeted women-related aspect, distinguish explicit from implicit harm, predict a fine-grained abuse category, and generate a supporting rationale. This motivates a unified four-task benchmark that connects target identification with harmfulness, categorization, and explanation.

Explainable and knowledge-guided reasoning. Recent approaches generate rationales through LLM distillation, multimodal debate, retrieval augmentation, or multimodal prompting (Lin et al., 2023, 2024, 2025; Hee and Lee, 2025). External and LMM-generated knowledge has also been used to improve implicit meme interpretation (Grasso et al., 2024; Tzelepi and Mezaris, 2025; Garg et al., 2025; Tripathi et al., 2026). Nevertheless, retrieved knowledge may be unrelated to the actual target, and fluent rationales may not reflect the evidence responsible for the prediction. Neuro-symbolic methods offer a complementary direction by combining neural representations with structured knowledge and rules (Colelough and Regli, 2025; Liu et al., 2025; Wang et al., 2025). Motivated by these limitations, SHEHARM-CAR uses the predicted women-related aspect to retrieve ontology knowledge, contrasts harm-supporting rules with contextual exception rules, and applies neural–symbolic consistency and counterfactual learning to align predictions with target-specific evidence.

3 Methodology

We propose SHEHARM-CAR, a counterfactual, target-guided, and rule-grounded neuro-symbolic framework for explainable women-targeted harmful meme detection. Given a meme image and its OCR text, the model learns a multimodal representation, identifies the women-related target, retrieves target-relevant ontology concepts, contrasts harm-supporting and contextual exception rules, and jointly predicts harmfulness, harm category, and a grounded rationale. Figure 2 presents the overall architecture.

3.1 Task Formulation

SheHarm-Meme supports four tasks: women-related target identification, three-way harmfulness classification, five-way harm-category classification, and evidence-grounded rationale generation. The women-related target is predicted from the fused image–text representation over a controlled inventory of ontology-linked target concepts, allowing targets to be expressed visually, textually, or through both modalities. Related studies have identified social entities targeted by harmful memes, detected victims using textual and visual entity cues, and characterized entities as victims, villains, or heroes (Pramanick et al., 2021; Sharma et al., 2022, 2023). Unlike these broader entity-level settings, our task focuses specifically on the woman, female role, body-related reference, relationship, profession, or women-associated group toward which the meme's meaning is directed.

Let

$$\mathcal{D} = \{(I_i, T_i, t_i, y_i, c_i, r_i)\}_{i=1}^{N}, \tag{1}$$

where I_i and T_i denote the meme image and OCR text, t_i is the women-related target label, y_i is the harmfulness label, c_i is the harm category, and r_i is the evidence-grounded rationale. The model jointly predicts

$$f_\theta(I_i, T_i) \rightarrow (\hat{t}_i, \hat{y}_i, \hat{c}_i, \hat{r}_i). \tag{2}$$

The harmfulness labels are *Explicit-Harm*, *Implicit-Harm*, and *Non-Harm*. Harmful memes receive one of five categories: *Sexual-Harassment*, *Violence*, *Misogyny*, *Appearance-Attack*, or *Character-Assassination*. For *Non-Harm* instances, $c_i =$ *None* and the category loss is masked.

3.2 Multimodal and Target Representation

Pretrained visual and textual encoders produce patch-level and token-level representations from the meme image and OCR text. Bidirectional cross-modal attention allows OCR tokens to attend to visual regions and image patches to attend to textual cues. The pooled outputs are fused into a global multimodal representation $\mathbf{z}_i$.

Unlike textual aspect extraction, the women-related target may be expressed visually, textually, or through their interaction. We therefore formulate target identification as classification over a controlled inventory of ontology-linked women-related target concepts. The target distribution is predicted as

$$\mathbf{p}_i^t = \mathrm{softmax}\left(\mathbf{W}_t \mathbf{z}_i + \mathbf{b}_t\right), \tag{3}$$

where each class corresponds to a canonical target concept, such as *woman passenger*, *female politician*, *woman's face*, or *women as a group*. The target classifier is supervised using cross-entropy:

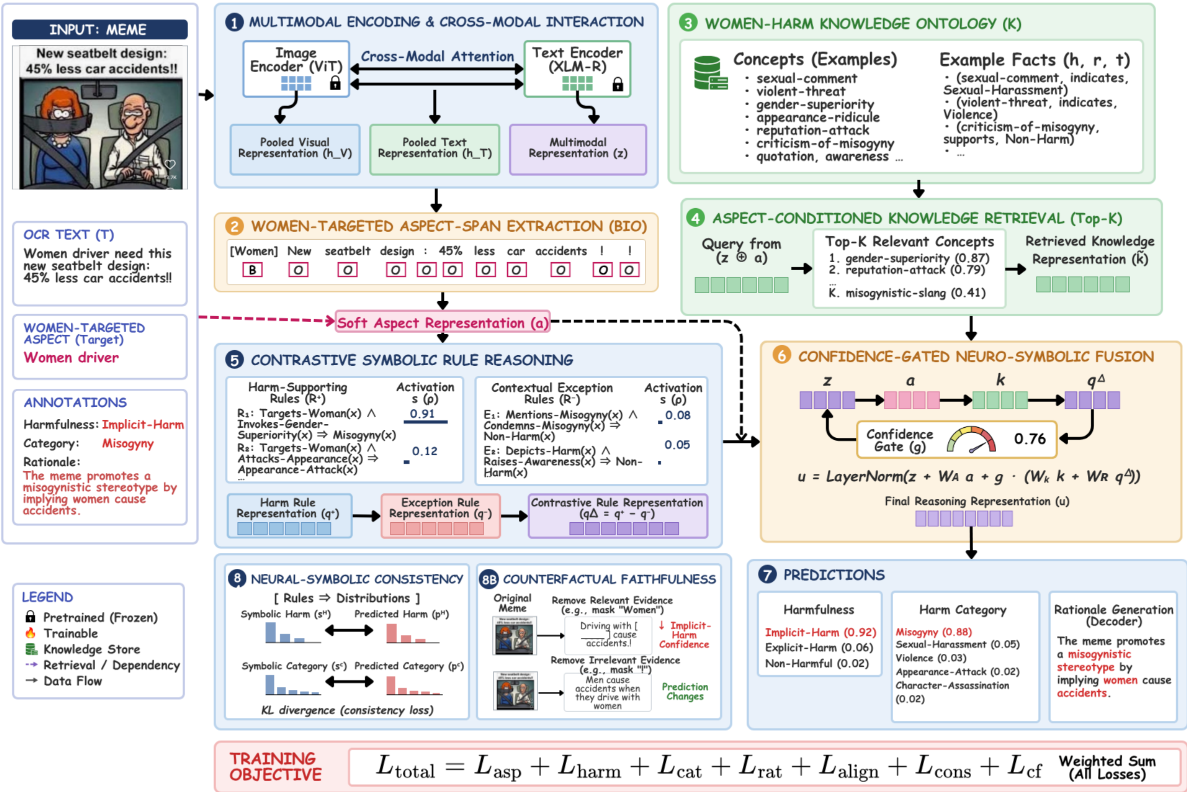

Figure 2: Architecture of SHEHARM-CAR. The framework combines women-targeted aspect extraction, aspect-conditioned ontology retrieval, contrastive symbolic reasoning, confidence-gated neural–symbolic fusion, and counterfactual faithfulness learning.

$$\mathcal{L}_{\text{tgt}} = -\sum_{i=1}^{N} \log p^{t}_{i,t_i}, \tag{4}$$

where t_i denotes the gold target label.

To preserve prediction uncertainty and enable differentiable target-conditioned reasoning, we compute a soft target representation:

$$\tilde{\mathbf{t}}_i = \sum_{m=1}^{M} p^{t}_{im}\mathbf{e}^{t}_{m}, \tag{5}$$

where M is the number of target concepts and $\mathbf{e}^t_m$ is the embedding of target concept m. Downstream retrieval, rule reasoning, classification, and rationale generation modules use $\tilde{\mathbf{t}}_i$ during both training and inference.

3.3 Target-Conditioned Ontology Retrieval

We construct a Women-Harm Knowledge Ontology containing women-related targets, harmful expressions, harm categories, contextual exceptions, and multimodal evidence. Each knowledge item is represented as a triple $\langle h, r, t\rangle$, where h, r, and t denote the head entity, relation, and tail entity, respectively. Table 1 presents representative ontology relations covering women-related targets, harm cues, contextual exceptions, and multimodal evidence. The ontology contains 611 concepts, 14 relation types, 1,287 triples, and 36 soft reasoning rules; the complete inventory is provided in the appendix. Table 14 summarizes the ontology statistics.

The retrieval query is conditioned on the global meme representation and the predicted soft target representation:

$$\mathbf{q}^K_i = \text{LayerNorm}\left(\mathbf{W}_q\left(\mathbf{z}_i \,\|\, \tilde{\mathbf{t}}_i\right) + \mathbf{b}_q\right). \tag{6}$$

The top-K ontology concepts are selected using cosine similarity between $\mathbf{q}^K_i$ and each ontology concept embedding $\mathbf{k}_u$. The retrieved knowledge representation is

$$\tilde{\mathbf{k}}_i = \sum_{u \in \mathcal{N}_K(i)} \alpha_{iu}\mathbf{k}_u, \tag{7}$$

where

$$\alpha_{iu} = \frac{\exp(s_{iu}/\tau)}{\sum_{v\in\mathcal{N}_K(i)} \exp(s_{iv}/\tau)}, \tag{8}$$

Type	Head	Relation / Tail	Reasoning Role
Target	woman passenger	*is-a* Female-Role	Identifies the women-related target.
Target	her face	*is-a* Female-Appearance	Grounds an appearance-related target.
Harm cue	"too ugly"	*expresses* Appearance-Attack	Links appearance ridicule to its category.
Harm cue	"beat her"	*expresses* Violence	Encodes physical-harm evidence.
Harm cue	"belongs in the kitchen"	*expresses* Misogyny	Represents a gender-role stereotype.
Context	harmful statement	*negated-by* Condemnation	Reduces harm support under criticism.
Context	harmful phrase	*quoted-in* Counter-Speech	Distinguishes quotation from endorsement.
Evidence	Violence	*supported-by* Visual Injury	Connects a category to visual evidence.

Table 1: Representative concepts and relations from the Women-Harm Knowledge Ontology. Contextual relations encode exceptions that distinguish harmful endorsement from quotation, criticism, and counter-speech.

s_{iu} is the cosine similarity between the query and concept u, and τ is a temperature parameter. A contrastive alignment loss $\mathcal{L}_{\text{align}}$ brings the target-conditioned query closer to relevant ontology concepts while separating it from sampled negative concepts.

3.4 Contrastive Symbolic Reasoning

Table 2 presents representative soft rules for harm-category prediction, harmfulness reasoning, contextual exception handling, and confidence control. The rule inventory contains harm-supporting rules $\mathcal{R}^+$ and contextual exception rules $\mathcal{R}^-$. For example, a women-related target combined with appearance ridicule supports *Appearance-Attack*, whereas an otherwise harmful expression that is explicitly condemned supports *Non-Harm*. The rules contribute graded evidence rather than hard decisions.

Visual relevance is estimated using the cross-modal attention assigned to image patches, while target-indicative OCR tokens are identified from text-to-image attention. High-scoring regions and tokens provide target-relevant evidence, whereas low-attention regions are used for irrelevant-evidence interventions.

Rule	Description
R1	IF women-related target AND sexualized remark or objectification THEN *Sexual-Harassment*.
R2	IF women-related target AND threat, violent intent, or encouragement of physical harm THEN *Violence*.
R3	IF women-related target AND gender-based inferiority, exclusion, or stereotype THEN *Misogyny*.
R4	IF women-related target AND ridicule of face, body, clothing, or physical appearance THEN *Appearance-Attack*.
R5	IF women-related target AND reputation attack, moral judgement, or derogatory character allegation THEN *Character-Assassination*.
R6	IF a harmful cue directly refers to the identified women-related target THEN increase *Explicit-Harm* support.
R7	IF the harmful interpretation requires image–text association, cultural knowledge, sarcasm, or indirect implication THEN increase *Implicit-Harm* support.
R8	IF a harmful expression is quoted or depicted AND explicitly criticized, condemned, or challenged THEN support *Non-Harm*.
R9	IF the meme raises awareness, provides counter-speech, or reports harm without endorsement THEN support *Non-Harm*.
R10	IF image, OCR text, retrieved knowledge, and activated rules conflict THEN reduce the neural–symbolic confidence gate.

Table 2: Representative harm-supporting, exception, and confidence-control rules.

For rule r_j, predicate truth values are estimated from the global multimodal representation $\mathbf{z}_i$, soft target representation $\widetilde{\mathbf{t}}_i$, and retrieved knowledge $\widetilde{\mathbf{k}}_i$. Its activation is computed using a soft conjunction:

$$\rho_{ij} = \prod_{l=1}^{m_j} p_{ijl}, \tag{9}$$

where p_{ijl} is the estimated truth value of predicate l in rule r_j. The contrastive symbolic representation is

$$\mathbf{q}_i^{\Delta} = \sum_{r_j \in \mathcal{R}^+} \rho_{ij} \mathbf{e}_{r_j} - \sum_{r_j \in \mathcal{R}^-} \rho_{ij} \mathbf{e}_{r_j}, \tag{10}$$

where $\mathbf{e}_{r_j}$ denotes the embedding of rule r_j. This contrastive formulation distinguishes harmful endorsement from quotation, criticism, awareness, and counter-speech. The complete ontology and rule inventory are provided in the appendix.

3.5 Confidence-Gated Prediction and Rationale Generation

Because retrieved knowledge and symbolic rules may be uncertain, a confidence gate controls their contribution:

$$\gamma_i = \sigma\left(\mathbf{w}_g^\top\left(\mathbf{z}_i \,\|\, \widetilde{\mathbf{t}}_i \,\|\, \widetilde{\mathbf{k}}_i \,\|\, \mathbf{q}_i^\Delta\right) + b_g\right). \quad (11)$$

The ontology and symbolic representations are combined as

$$\mathbf{v}_i = \mathbf{W}_K\widetilde{\mathbf{k}}_i + \mathbf{W}_R\mathbf{q}_i^\Delta, \quad (12)$$

and the final reasoning representation is

$$\mathbf{u}_i = \mathrm{LayerNorm}\left(\mathbf{z}_i + \mathbf{W}_T\widetilde{\mathbf{t}}_i + \gamma_i\mathbf{v}_i\right). \quad (13)$$

Separate prediction heads produce the harmfulness and harm-category distributions:

$$\mathbf{p}_i^y = \mathrm{softmax}\left(\mathbf{W}_y\mathbf{u}_i + \mathbf{b}_y\right), \qquad \mathbf{p}_i^c = \mathrm{softmax}\left(\mathbf{W}_c\mathbf{u}_i + \mathbf{b}_c\right). \quad (14)$$

The harmfulness loss is

$$\mathcal{L}_{\mathrm{harm}} = -\sum_{i=1}^{N}\log p_{i,y_i}^y. \quad (15)$$

Because *Non-Harm* memes do not receive a harm category, the category loss is masked:

$$\mathcal{L}_{\mathrm{cat}} = -\sum_{i=1}^{N} m_i \log p_{i,c_i}^c, \quad (16)$$

where

$$m_i = \begin{cases} 1, & y_i \neq \textit{Non-Harm}, \\ 0, & y_i = \textit{Non-Harm}. \end{cases} \quad (17)$$

The rationale decoder is conditioned on $\mathbf{u}_i$, the predicted soft target representation, retrieved knowledge, and activated rules. It is trained using autoregressive cross-entropy $\mathcal{L}_{\mathrm{rat}}$ to generate a concise explanation that identifies the target, relevant multimodal evidence, harmfulness, and harm category.

3.6 Consistency and Counterfactual Faithfulness

Activated rules are mapped to symbolic harmfulness and category distributions, denoted by $\mathbf{s}_i^y$ and $\mathbf{s}_i^c$. Their agreement with the neural predictions is encouraged through

$$\mathcal{L}_{\mathrm{cons}} = \sum_{i=1}^{N} D_{\mathrm{KL}}\left(\mathbf{s}_i^y \| \mathbf{p}_i^y\right) + \sum_{i=1}^{N} m_i D_{\mathrm{KL}}\left(\mathbf{s}_i^c \| \mathbf{p}_i^c\right). \quad (18)$$

The consistency loss is applied only when at least one symbolic rule exceeds a validation-selected activation threshold, thereby limiting the influence of uncertain rule evidence.

We further construct evidence-based counterfactuals by suppressing target-relevant image regions or OCR cues, replacing the predicted target representation with a null representation, removing the highest-ranked ontology concept, or deactivating the strongest rule. Removing relevant evidence should reduce confidence in the original prediction, whereas perturbing low-relevance evidence should preserve it:

$$\mathcal{L}_{\mathrm{cf}} = \mathcal{L}_{\mathrm{nec}} + \lambda_{\mathrm{inv}}\mathcal{L}_{\mathrm{inv}}. \quad (19)$$

Here, $\mathcal{L}_{\mathrm{nec}}$ encourages a confidence decrease after relevant-evidence removal, while $\mathcal{L}_{\mathrm{inv}}$ encourages prediction invariance under irrelevant-evidence perturbation.

3.7 Joint Training and Inference

The supervised four-task objective is

$$\mathcal{L}_{\mathrm{task}} = \mathcal{L}_{\mathrm{tgt}} + \lambda_{\mathrm{harm}}\mathcal{L}_{\mathrm{harm}} + \lambda_{\mathrm{cat}}\mathcal{L}_{\mathrm{cat}} + \lambda_{\mathrm{rat}}\mathcal{L}_{\mathrm{rat}}. \quad (20)$$

The complete training objective is

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{task}} + \lambda_{\mathrm{align}}\mathcal{L}_{\mathrm{align}} + \lambda_{\mathrm{cons}}\mathcal{L}_{\mathrm{cons}} + \lambda_{\mathrm{cf}}\mathcal{L}_{\mathrm{cf}}. \quad (21)$$

The coefficients λ_{harm}, λ_{cat}, λ_{rat}, λ_{align}, λ_{cons}, and λ_{cf} control the respective loss contributions.

At inference, only the meme image and OCR text are provided. The model predicts the women-related target, retrieves target-relevant ontology concepts, activates harm-supporting and contextual exception rules, and outputs the harmfulness label, harm category, and evidence-grounded rationale. No gold targets, harmfulness labels, harm categories, or rationales are used at inference.

Split	Exp.	Imp.	Non.	Total
Train	1,498	877	547	2,922
Dev	187	109	68	364
Test	188	110	68	366
Total	**1,873**	**1,096**	**683**	**3,652**

Table 3: SheHarm-Meme split and harmfulness distribution. Exp., Imp., and Non. denote *Explicit-Harm*, *Implicit-Harm*, and *Non-Harm*, respectively.

Annotation Component	Agreement
Women-related Target Cohen's κ	0.81
Harmfulness Cohen's κ	0.76
Harm Category Cohen's κ	0.74
Rationale Token-F1	74.9

Table 4: Inter-annotator agreement before expert adjudication. Women-related target, harmfulness, and harm-category labels are evaluated using Cohen's κ, while rationales use token-level F1 (%).

4 Experimental Setup

4.1 Dataset

We initially collected 4,086 candidate memes from publicly accessible social-media platforms, including Reddit, Facebook, Instagram, X, and public discussion forums. Preprocessing removed 434 instances (10.62%) that were irrelevant to women-targeted harm, duplicated or near-duplicated, unreadable, privacy-sensitive, or lacked sufficient image–text context, resulting in 3,652 retained memes. Usernames, personal identifiers, and other traceable details were removed or anonymized before annotation.

SheHarm-Meme contains 3,652 memes annotated with women-related aspect spans, harmfulness labels, harm categories, and textual rationales. The harmfulness distribution comprises 1,873 *Explicit-Harm*, 1,096 *Implicit-Harm*, and 683 *Non-Harm* instances. Harmful memes are assigned one of five harm categories, while *Non-Harm* instances receive the null category *None*.

We partition the dataset into training, validation, and test sets using an approximately 80/10/10 split, resulting in 2,922, 364, and 366 instances, respectively. Stratification is performed jointly over the harmfulness and harm-category labels. Duplicate and near-duplicate memes are kept within the same split to prevent information leakage. All reported results are averaged over three random seeds. Table 3 presents the split-wise label distribution.

4.2 Annotation Schema and Guidelines

Each meme is independently annotated by two trained undergraduate students aged 19–21 who are proficient in the dataset language and familiar with informal social-media discourse. They annotate four components: the women-related aspect span, harmfulness label, harm category, and textual rationale. The aspect is the shortest meaningful OCR span referring to the targeted woman, female role, relationship, profession, appearance, behaviour, or another women-related expression.

Harmfulness is labeled as *Explicit-Harm*, *Implicit-Harm*, or *Non-Harm*. Harmful memes receive one of five harm categories, whereas non-harmful memes are assigned *None*. Rationales identify the target-specific image–text evidence and briefly justify the harmful or non-harmful interpretation.

Annotators receive written guidelines, example cases, and content warnings, and may skip distressing instances. Difficult cases include sarcasm, indirect image–text associations, quotation, counter-speech, culturally grounded references, and noisy OCR. Disagreements are resolved through discussion and expert adjudication. Table 4 reports agreement before adjudication.

4.3 Evaluation Metrics

We evaluate aspect-span extraction using token-level F1 and exact-span F1. Harmfulness and harm-category classification are evaluated using macro-F1, with harm-category performance computed only over harmful instances. We additionally report joint tuple F1 for correctly predicting the aspect, harmfulness label, and harm-category.

Rationale quality is evaluated using BERTScore (Zhang et al., 2020) and human assessment of relevance, faithfulness, and informativeness. Counterfactual faithfulness is measured by the confidence decrease after removing target-relevant evidence and the prediction stability after removing low-relevance evidence.

4.4 Baselines

We compare SheHarm-CAR with text-only RoBERTa (Liu et al., 2019); multimodal encoders ViLT (Kim et al., 2021) and Hate-CLIPper (Kumar and Nandakumar, 2022); vision–language models LLaVA (Liu et al., 2023), InternVL (Chen et al., 2024), Llama-3.2-Vision-11B, and Qwen2.5-VL-7B (Qwen Team, 2025); and knowledge-, interpretation-, or reasoning-guided methods KERMIT (Grasso et al., 2024), KID-VLM (Garg et al., 2025), IntMeme$_{\text{INSTRUCTBLIP}}$ (Hee and Lee, 2025), ExplainHM++ (Lin et al., 2025), and SGoT-R1 (Wang et al., 2026). Encoder-based methods use the same task-specific heads where required, while prompted models follow a fixed output schema and identical label definitions.

4.5 Implementation Details

We implement SHEHARM-CAR in PyTorch. CLIP ViT-B/32 is used as the visual encoder, while RoBERTa-base encodes the OCR text. Images are resized to 224×224, and OCR sequences are truncated to 128 tokens. The multimodal, ontology, and symbolic representations have a hidden dimension of 768. Bidirectional cross-modal attention, the confidence gate, and newly initialized prediction layers use a dropout rate of 0.2. For aspect-conditioned retrieval, the model selects the top-$K = 5$ ontology concepts using cosine similarity with temperature $\tau = 0.07$. Ten hard-negative concepts are sampled for the alignment objective, and rules with activation scores above a validation-selected threshold of 0.60 contribute to symbolic reasoning. The rationale decoder uses beam search with beam size 4 and a maximum output length of 64 tokens. The model is optimized using AdamW with learning rates of 1×10^{-5} for the pretrained encoders and 5×10^{-4} for newly initialized layers, together with a weight decay of 0.01. Training uses a batch size of 32 for up to 15 epochs, mixed-precision computation, and early stopping with a patience of five epochs. The checkpoint with the highest mean validation F1 across aspect extraction, harmfulness classification, harm-category classification, and rationale generation is retained. Experiments are conducted on a single NVIDIA A100 GPU, and results are reported as the mean over three random seeds. We set the aspect-loss coefficient implicitly to 1.0. The supervised loss weights are $\lambda_{\text{harm}} = \lambda_{\text{cat}} = \lambda_{\text{rat}} = 1.0$, while $\lambda_{\text{align}} = 0.1$, $\lambda_{\text{cons}} = 0.1$, and $\lambda_{\text{cf}} = 0.2$. Within the counterfactual objective, λ_{inv} is set to 0.5. All coefficients are selected using validation performance.

5 Results and Analysis

SHEHARM-CAR achieves the strongest performance across all metrics. Compared with SGoT-R1, the strongest baseline, it improves Harm-F1 and Joint-F1 by 2.04 and 2.59 points, respectively. The gain in Joint-F1 indicates more consistent prediction of the target, harmfulness, and harm category, while the improvements in BERTScore and CF-Faith. suggest better grounded rationale generation. Table 7 shows that every proposed component contributes to performance. Removing aspect conditioning produces the largest Joint-F1 decline, while removing exception rules particularly weakens harmfulness discrimination. The counterfactual objective has the strongest effect on CF-Faith., supporting its role in aligning rationales with decision-relevant evidence.

Setting	Value
Encoders	CLIP ViT-B/32; RoBERTa-base
Input size	224×224; 128 OCR tokens
Hidden size / dropout	768 / 0.2
Ontology retrieval	$K = 5$, $\tau = 0.07$, 10 hard negatives
Rule threshold	0.60
Encoder learning rate	1×10^{-5}
New-layer learning rate	5×10^{-4}
Weight decay	0.01
Batch / epochs / patience	32 / 15 / 5
Rationale decoding	Beam size 4; maximum 64 tokens
Task-loss weights	$\lambda_{\text{harm}} = 1.0$, $\lambda_{\text{cat}} = 1.0$, $\lambda_{\text{rat}} = 1.0$
Auxiliary-loss weights	$\lambda_{\text{align}} = 0.1$, $\lambda_{\text{cons}} = 0.1$, $\lambda_{\text{cf}} = 0.2$
Counterfactual invariance	$\lambda_{\text{inv}} = 0.5$
Optimizer / precision	AdamW / mixed precision
Runs / hardware	3 seeds / one NVIDIA A100 GPU

Table 5: Hyperparameter settings for SHEHARM-CAR.

5.1 Main Results

Table 6 compares the proposed framework with unimodal, multimodal, knowledge-guided, and explainable baselines. We report performance on all four tasks together with joint structured prediction and rationale-faithfulness measures.

5.2 Ablation Study

We remove each major component to assess its contribution: aspect conditioning, ontology retrieval, contrastive exception rules, confidence gating, neural–symbolic consistency, and counterfactual learning.

5.3 Class-Wise Harmfulness Analysis

Table 8 reports class-wise harmfulness performance. SHEHARM-CAR achieves the strongest results for all three classes, with the largest advantage on *Implicit-Harm*. This suggests that aspect-

Model	Tgt.-F1	Harm-F1	Cat.-F1	Joint-F1	BERTScore	CF-Faith.
Text-only RoBERTa (Liu et al., 2019)	65.40	71.20	68.35	59.74	79.10	70.2
ViLT (Kim et al., 2021)	69.82	75.91	72.63	64.35	81.84	73.8
Hate-CLIPper (Kumar and Nandakumar, 2022)	73.48	78.56	75.91	67.95	83.41	76.4
LLaVA (Liu et al., 2023)	74.61	79.34	76.82	68.91	84.52	77.6
InternVL (Chen et al., 2024)	76.18	80.47	78.06	70.35	85.39	78.9
Llama-3.2-Vision-11B	76.94	80.91	78.72	71.06	85.86	79.5
Qwen2.5-VL-7B (Qwen Team, 2025)	78.26	81.64	79.55	72.18	86.63	80.7
KERMIT (Grasso et al., 2024)	78.94	81.10	79.36	72.28	86.05	80.8
KID-VLM (Garg et al., 2025)	79.31	82.03	80.14	73.02	87.02	81.5
IntMeme$_{\text{INSTRUCTBLIP}}$ (Hee and Lee, 2025)	79.74	82.31	80.26	73.28	87.42	81.9
ExplainHM++ (Lin et al., 2025)	80.12	81.96	80.48	73.64	87.14	82.2
SGoT-R1 (Wang et al., 2026)	80.74	82.58	81.16	74.29	87.83	83.0
SheHarm-CAR	**82.05**	**84.62**	**82.91**	**76.88**	**89.34**	**85.6**

Table 6: Comparison on SheHarm-Meme. All baselines are evaluated using the same dataset splits and task setting. Tgt.-F1 denotes macro-F1 for women-related target identification; category macro-F1 is computed only over harmful instances. CF-Faith. denotes counterfactual faithfulness.

Variant	Harm-F1	Joint-F1	CF-Faith.
w/o Target Conditioning	81.74	72.95	81.3
w/o Ontology Retrieval	82.16	73.84	82.0
w/o Exception Rules	81.88	73.42	80.8
w/o Confidence Gate	82.73	74.61	82.7
w/o $\mathcal{L}_{\text{cons}}$	83.11	75.24	81.9
w/o $\mathcal{L}_{\text{cf}}$	83.54	75.72	78.4
Full Model	**84.62**	**76.88**	**85.6**

Table 7: Ablation results for SHEHARM-CAR.

Model	Expl.-F1	Impl.-F1	Non-Hm.-F1
ViLT	80.6	68.9	77.2
Hate-CLIPper	82.1	71.4	78.6
Qwen2.5-VL-7B	84.0	74.8	80.7
KID-VLM	84.7	76.1	81.5
ExplainHM++	85.2	77.0	82.3
SGoT-R1	85.8	78.1	82.9
SheHarm-CAR	**87.3**	**80.4**	**84.1**

Table 8: Class-wise harmfulness performance on SheHarm-Meme. Expl.-F1, Impl.-F1, and Non-Hm.-F1 denote F1 scores for *Explicit-Harm*, *Implicit-Harm*, and *Non-Harm*, respectively.

Intervention	Δ Conf.	Flip Rate
Suppress target	18.6	27.4
Mask relevant region	15.2	22.8
Remove top concept	10.8	15.6
Deactivate strongest rule	12.4	18.1
Mask irrelevant region	2.1	3.6

Table 9: Counterfactual evidence analysis.

Model	CFR↓	FPR-H↓	RS↑
ViLT	14.2	17.1	64.1
Hate-CLIPper	12.9	15.6	67.3
Qwen2.5-VL-7B	10.7	13.2	71.4
KID-VLM	9.8	12.1	73.2
ExplainHM++	9.2	11.4	74.6
SGoT-R1	8.7	10.8	75.3
w/o Exception Rules	8.3	10.2	73.8
w/o $\mathcal{L}_{\text{cf}}$	7.9	9.8	71.5
SheHarm-CAR	**6.8**	**8.7**	**78.1**

Table 10: Robustness under lexical counterfactuals. CFR: counterfactual flip rate; FPR-H: false-positive rate on non-harmful counterfactuals; RS: rationale stability.

conditioned ontology retrieval and contrastive rule reasoning are particularly useful when harm is not expressed through direct lexical cues.

We further evaluate faithfulness through evidence masking. Table 9 reports the average confidence decrease and prediction-flip rate after masking the predicted aspect, the most relevant visual region, and an irrelevant region. Larger changes for target-relevant interventions, together with stability under irrelevant masking, indicate stronger reliance on evidence connected to the model's prediction.

5.4 Counterfactual Robustness

Table 10 evaluates robustness under counterfactual lexical substitutions. SHEHARM-CAR obtains the lowest counterfactual flip rate and false-positive rate while maintaining the highest rationale stability. Removing exception rules increases false harmful predictions, whereas removing $\mathcal{L}_{\text{cf}}$ most strongly reduces rationale stability. These findings suggest that contextual rules and counterfactual learning reduce dependence on isolated lexical cues.

5.5 Evaluation on Public Benchmarks

We further evaluate SHEHARM-CAR on Facebook Hateful Memes (FHM) (Kiela et al., 2020), Multimedia Automatic Misogyny Identification (MAMI) (Fersini et al., 2022), and HarMeme (Pramanick et al., 2021). These benchmarks cover multimodal hate, misogyny, and harmful-meme detection, respectively. As they do not provide the complete target, category, and rationale annotations of SheHarm-Meme, we retain the multimodal, ontology, rule-reasoning, and confidence-gated components and optimize only the supported classification objectives. Table 11 reports AUROC and accuracy using official or matched splits.

5.6 Human Evaluation

Three evaluators independently assess 300 generated rationales on a five-point Likert scale for target relevance, evidence grounding, informativeness, and faithfulness. The evaluation comprises rationales generated by the three compared models for the same 100 test instances, sampled to cover explicit, implicit, and non-harmful memes as well as the five harm categories. Rationales are presented in randomized order without revealing model identities. We report the mean score for each criterion and measure inter-rater agreement using Krippendorff's α. The evaluators achieve an overall Krippendorff's α of 0.79, indicating substantial agreement. Table 12 summarizes the results.

6 Conclusion

We introduced SheHarm-Meme, a four-task benchmark for women-targeted harmful meme understanding, and SHEHARM-CAR, an aspect-guided neuro-symbolic framework. The approach connects women-related aspect extraction with harmfulness classification, harm categorization, and grounded rationale generation. Aspect-conditioned ontology retrieval, contrastive harm and exception rules, and confidence-gated fusion support implicit reasoning, while consistency and counterfactual objectives encourage faithful explanations. SHEHARM-CAR improves Harm-F1 and Joint-F1 over the strongest baseline and demonstrates greater stability under evidence and lexical counterfactuals. Future work will extend the ontology across languages and cultural contexts and investigate visual target localization beyond OCR-based aspect spans.

Limitations

SheHarm-Meme represents only the languages, cultures, platforms, and harm categories included during data collection, and may not capture all forms of women-targeted abuse. Some labels, particularly implicit harm and contextual exceptions, require subjective social and cultural interpretation. The current aspect task primarily identifies OCR spans and therefore does not fully localize visually expressed targets. The ontology and rule inventory may also contain incomplete or context-dependent knowledge. Finally, counterfactual masking provides an approximate measure of faithfulness and does not establish causal correctness in all cases.

Ethics Statement

SheHarm-Meme contains potentially offensive and gender-targeted content. Data collection and release follow the terms and access conditions of the source platforms. Personally identifying information is removed, and meme authors are not profiled or identified. Annotators receive clear content warnings, may skip distressing examples, and are provided with regular breaks and support procedures.

The benchmark is intended solely for research on harm detection, moderation support, and explainable AI. It should not be used for profiling individuals or making punitive decisions without human review. Because harmfulness depends on social and cultural context, model outputs may reproduce biases present in the data, pretrained encoders, ontology, or annotation process. We therefore recommend human oversight, subgroup evaluation, and cautious deployment.

Acknowledgments

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Model	FHM		MAMI		HarMeme	
	AUROC	Acc.	AUROC	Acc.	AUROC	Acc.
Hate-CLIPper (Kumar and Nandakumar, 2022)	78.64	69.85	79.72	70.94	87.16	81.28
KID-VLM (Garg et al., 2025)	80.27	71.16	80.83	71.65	88.42	82.14
IntMeme$_{\text{INSTRUCTBLIP}}$ (Hee and Lee, 2025)	81.05	71.48	81.59	72.44	88.00	82.66
SGoT-R1 (Wang et al., 2026)	82.14	72.36	82.47	73.08	89.54	83.31
SheHarm-CAR	**83.46**	**73.72**	**83.81**	**74.29**	**90.76**	**84.58**

Table 11: Cross-dataset evaluation on public harmful-meme benchmarks. All models are trained and evaluated using the official or matched dataset splits. For datasets without women-related target and rationale annotations, only the classification objectives supported by the corresponding benchmark are optimized.

Model	Rel.	Ground.	Info.	Faith.
ExplainHM++	3.82	3.69	3.76	3.61
SGoT-R1	3.96	3.84	3.89	3.78
SheHarm-CAR	**4.21**	**4.13**	**4.08**	**4.16**

Table 12: Human evaluation of generated rationales on a five-point scale. Rel., Ground., Info., and Faith. denote relevance, groundedness, informativeness, and faithfulness, respectively.

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A Additional Experimental Details

Table 15 highlights recurring challenges. Implicit stereotypes and humour can be mistaken for non-harm, while salient visual violence may overshadow the underlying misogynistic intent. Lexically sensitive models may also assign harm based on isolated words without considering their benign contextual use. In contrast, target-conditioned ontology retrieval and exception rules support correct decisions when the target and communicative intent are sufficiently grounded across modalities.

Relation Type	Head Entity	Relation	Tail Entity	Reasoning Role
Target relation	*wife*	`is_women_related_role`	*Female-Relationship*	Identifies a women-related target or aspect.
Target relation	*female politician*	`is_women_related_role`	*Female-Profession*	Grounds the target in a professional role.
Harm cue	*belongs in the kitchen*	`expresses`	*Misogyny*	Associates a stereotypical statement with a harm category.
Harm cue	*too ugly*	`expresses`	*Appearance-Attack*	Links an appearance-directed insult to its category.
Harm cue	*she is characterless*	`expresses`	*Character-Assassination*	Represents reputation- or morality-based attacks.
Harm cue	*rape threat*	`expresses`	*Sexual-Harassment*	Grounds sexually threatening content.
Harm cue	*beat her*	`expresses`	*Violence*	Connects physical-harm expressions with violence.
Context relation	*harmful phrase*	`quoted_in`	*Counter-Speech*	Prevents quoted harmful language from being treated as endorsement.
Context relation	*stereotypical statement*	`negated_by`	*Negation Cue*	Reduces harm support when the statement is explicitly rejected.
Evidence relation	*Misogyny*	`supported_by`	*Textual or Visual Evidence*	Connects the predicted category with multimodal evidence.

Table 13: Representative triples from the Women-Harm Knowledge Ontology. Contextual relations encode exceptions that distinguish harm endorsement from quotation, negation, and counter-speech.

Ontology Component	Count
Women-related target concepts	126
Harm-cue concepts	438
Harm-category concepts	5
Context and exception concepts	42
Relation types	14
Ontology triples	1,287
Soft reasoning rules	36

Table 14: Statistics of the Women-Harm Knowledge Ontology.

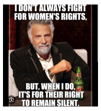
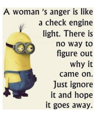
ID	Meme	Target	Gold	Prediction	Error Type	Qualitative Analysis
Q1		women as a group	*Implicit-Harm* *Misogyny*	*Non-Harm* *None*	Implicit stereotype missed	The harmful meaning depends on associating women's preferences with wealth, cars, and housing. Without an explicit insult, the model interprets the comparison as harmless humour.
Q2		woman being struck	*Explicit-Harm* *Misogyny*	*Explicit-Harm* *Violence*	Competing categories	The model detects physical aggression but prioritizes the depicted slap over the central kitchen stereotype. This illustrates ambiguity when violence reinforces misogynistic role expectations.
Q3		women as a group	*Implicit-Harm* *Misogyny*	*Non-Harm* *None*	Conditional harm missed	The statement initially appears supportive of women but subsequently conditions respect on satisfying unspecified gendered expectations. The model overweights the positive phrase "respect a woman."
Q4		women athletes	*Explicit-Harm* *Misogyny*	*Explicit-Harm* *Appearance-Attack*	Category confusion	The model correctly detects explicit harm but misclassifies the attack because it attends to the women shown in the image rather than the textual denial of women's athletic ability and kitchen stereotype.
Q5		woman speaker	*Non-Harm* *None*	*Explicit-Harm* *Sexual-Harassment*	Lexical shortcut	The model over-relies on the phrase "cop porn" and predicts sexual harm, although the meme presents a benign phonetic misunderstanding without degrading or targeting the woman.
Q6		wife	*Explicit-Harm* *Appearance-Attack*	*Non-Harm* *None*	Humour masks harm	The conversational format obscures the appearance-based attack. The model fails to recognize that the husband's response uses the wife's weight as the basis of the joke.
Q7		women as a group	*Implicit-Harm* *Misogyny*	*Non-Harm* *None*	Analogy-based stereotype missed	The meme compares women's anger to an unexplained warning light and suggests that it should be ignored. The humorous analogy obscures the stereotype that women's emotions are irrational and not worth understanding.
Q8		woman soldier	*Implicit-Harm* *Misogyny*	*Non-Harm* *None*	Contradictory framing	The empowering visual and statement about joining the military mask the harmful implication that women invoke gender selectively to avoid equal treatment. The model therefore mistakes the stereotype for benign humour.

Table 15: Qualitative error analysis on SheHarm-Meme. Gold and predicted entries report harmfulness and harm category, respectively. Errors arise from implicit stereotypes, competing categories, conditional language, lexical shortcuts, humour, and image–text interaction.